PAPER • OPEN ACCESS

Research on camouflage target detection method based on improved YOLOv5

To cite this article: Wei Zhang *et al* 2022 *J. Phys.: Conf. Ser.* **2284** 012018

Research on camouflage target detection method based on improved YOLOv5

Wei Zhang[1], Qikai Zhou[1], Ruizhi Li[1] and Fu Niu[1*]

[1] Academy of Systems Engineering of Academy of Military Science, Beijing,100166, China

*Corresponding author's e-mail: zycver666@163.com

Abstract. Objective: In the task of camouflage target detection, there is a problem that the target is highly integrated with the complex environment background, which is difficult to identify and leads to false detection and missed detection. A target detection algorithm CM-YOLOv5s is proposed for camouflage characteristics. Method: The algorithm uses YOLOv5s as the basic framework. First, a coordinated attention mechanism is embedded in the backbone feature extraction network, which enhances the network's ability to extract camouflaged target features, weakens the attention to the surrounding background, and effectively improves the algorithm's anti-background. Interference ability; secondly, the Mixup data enhancement strategy is used to simulate overlapping occlusion scenarios, which further strengthens the network model's learning ability for complex samples. Results: The training and verification were carried out on the self-made Military Camouflage Target Dataset (MCTD), and the precision, recall, and average mAP of the improved CM-YOLOv5s algorithm reached 95.9%, 87.1%, and 94.1, respectively. %, compared with the original YOLOv5s model, the average accuracy rate is improved by 3.8 percentage points. Conclusion: The improved algorithm has better detection effect, and realizes accurate identification and rapid positioning of military camouflage targets in complex environments.

1.Introduction

In recent years, target detection technology has been widely used in all walks of life, especially in military and civilian applications. Especially with the rapid development of computer technologies such as artificial intelligence and deep learning, in the military field, modern warfare has developed from the previous mechanization to an unmanned, information-based and intelligent mode, and various cutting-edge weapons and equipment around the world are gradually becoming The development of science and technology and intelligence is perfect, especially military target detection technology has obviously become the key technology of battlefield situational awareness. For intelligent weapons and equipment, the first problem to be solved is to make them have keen observation and identification capabilities. Therefore, in the complex and changeable battlefield environment, it is of great strategic significance for us to automatically and quickly identify potential enemy military targets[1].

In the civil field, when a natural disaster strikes unexpectedly, it may cause a large area of houses to collapse and cause casualties, so the top priority is to rescue the survivors in the rubble in time. At this time, rescue robots and search and rescue drones play a major role in emergency rescue. Rescue robots can quickly identify and locate survivors and carry out rescue work in complex and changeable disaster sites, especially in narrow spaces where fire rescue personnel, search and rescue dogs or other rescue equipment are difficult to reach quickly; search and rescue drones Compared with traditional remote sensing images and high-altitude aerial photography technology, search and rescue drones with target recognition and positioning functions are more efficient. In the future, more advanced and mature target detection technology will be applied to equipment such as search and rescue robots or search and rescue drones, which can greatly improve the search and rescue efficiency of personnel.

At present, the existing target detection algorithms are not outstanding in the target detection effect in the above-mentioned complex environments, especially in the detection of camouflaged targets in complex environments, which are more difficult than conventional detection tasks. In addition, military camouflage technology mainly changes the surface characteristics of the target through colors, textures and various patterns, so as to achieve the effect of integrating into the surrounding complex environment, so as to achieve the purpose of stealth, and some of the camouflage effects even reach the level that the human eye cannot recognize [2]. Not only that, military targets are usually located in complex field environments, such as mountains, snow, deserts, and bushes, which may contain a variety of military targets, all of which bring great challenges to the existing target detection algorithms..

After research and analysis, the difficulty of camouflaged target detection mainly has the following two points: (1) The concealment of the target is very high and the recognition degree is very low. Due to the complex diversity of its own dyes, patterns and lines, the target is extremely integrated with the surrounding environment. Discrimination is low; (2) Overlapping occlusion is more common. Since the loss of target feature information is proportional to the degree of occlusion, overlapping occlusion will cause the algorithm network to lose part of the camouflage feature information during the feature extraction process, which in turn makes the training of the neural network more difficult. Therefore, if a convolutional neural algorithm with high accuracy and strong generalization ability can be developed, it will have important practical significance for the rapid search and positioning of targets in the field of intelligent combat and battlefield search and rescue.

At present, the algorithm research on military camouflage target detection is mainly divided into two types: traditional target detection algorithm and target detection algorithm based on deep learning. The traditional detection algorithm mainly detects the target by obtaining the structural information of the camouflage image. Since it only pays attention to the shallow feature information of the image itself, it is difficult to detect the targets with high concealment, different shapes and overlapping occlusions in complex environments. Not ideal, the commonly used traditional target detection algorithms include support vector machine SVM model [3], iterator AdaBoosting [4], DPM [5], random forest RF model [6] and so on. With the rapid development of artificial intelligence and graph neural network technology, as well as the substantial improvement of computer hardware performance, some scholars have applied deep learning-based target detection algorithms to military camouflage target detection tasks. The deep learning-based military camouflage target detection algorithms mainly It is divided into two categories: one is the two-stage target detection algorithm (Two-Stage), and its detection process is mainly divided into two steps: feature extraction and feature classification. The representative algorithms are R-CNN[7], Fast R-CNN[8], Faster R-CNN[9], etc.; the other is a one-stage target detection algorithm (One-Stage), which directly classifies and predicts the target on the feature map, and the representative algorithms include SSD[10], EfficienctNet[11], RetinaNet [12] and YOLO series [13-16], etc. The two-stage algorithm also has higher detection accuracy, while the single-stage algorithm has an advantage in inference speed. At present, many studies have applied target detection algorithms based on deep learning in military fields such as precision guidance and battlefield search and rescue, and have achieved good detection results. Wang Zhi et al. [17] detected and recognized the occlusion and camouflage of tank model images in complex backgrounds, and the detection accuracy of the improved convolutional neural network algorithm reached 82.6%. In view of the low detection accuracy of traditional detection algorithms, Yu Bowen et al. [18] further improved the detection accuracy of military targets by introducing methods such as improved ResNet50-D residual network and dual attention mechanism. Based on the RetinaNet algorithm, Deng Xiaotong et al. [19] introduced an attention mechanism to imitate human vision, and improved the detection accuracy of camouflage targets to 93.1%. The above research has achieved good results for the task of military camouflage target detection, but there is still room for improvement in detection accuracy.

To this end, based on the YOLOv5s algorithm framework, this paper adds Coordinate Attention (CA) [20] to the backbone network Backbone and feature fusion network Neck for the characteristics of camouflaged targets. By adding an attention mechanism, the network has The same keen observation ability as the human eye improves the network's extraction of feature information of military camouflage targets, and further strengthens the algorithm's anti-background interference ability; secondly, the Mixup [21] data augmentation strategy is adopted in the training process. This strategy can simulate overlapping occlusion scenarios and further strengthen the model's learning ability for complex samples. By making the above two improvements on the basis of the YOLOv5 algorithm, this paper proposes a target detection algorithm CA-Mixup-YOLOv5s (hereinafter referred to as "CM-YOLOv5s") for the camouflage characteristics of camouflage, which improves the detection accuracy of camouflaged targets in complex environments

2.Materials and Methods

This chapter first briefly outlines the basic principles of the YOLOv5 algorithm, and analyzes the shortcomings and deficiencies of the algorithm in the task of camouflaged target detection. For the detection task of camouflaged targets in complex environments, this study proposes two improvements: First, the CA coordination attention module is introduced into the backbone feature network of YOLOv5, and the attention mechanism is used to strengthen the network's attention to camouflaged targets and improve the network's awareness of the target content. The ability to extract feature information such as direction and position, weaken complex background information, and improve the algorithm's accurate detection and recognition of camouflaged targets. The second is to use the Mixup method to simulate occlusion and overlapping scenes in camouflaged images in complex environments, enhance the network's ability to learn complex samples, and further improve the generalization ability of the network model.

2.1 The basic principle of YOLOv5 algorithm

The YOLOv5 algorithm [22] is a detection algorithm with excellent detection speed and detection accuracy. The algorithm consists of four parts: the input, the backbone, the neck, and the prediction. Its detection framework is shown in Figure 1.

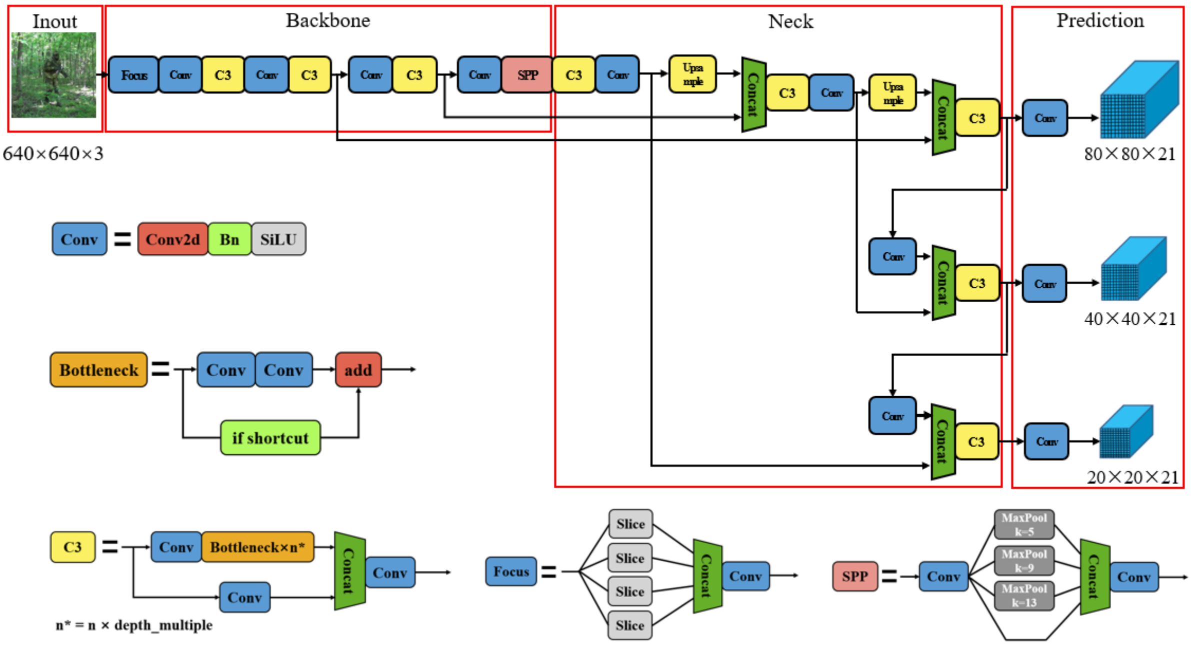

Figure 1. YOLOv5s detection framework

The first part of the YOLOv5 algorithm is the input, and the size of the input image is generally 640×640. The second part is the backbone feature extraction network, which is mainly composed of

Focus, Conv, C3, and Spatial Pyramid Pooling (SPP) [23] and other structures. The Focus structure is to slice and splicing the input feature map. In order to retain more feature information, Conv is the basic convolution process, which mainly performs three operations such as 2-dimensional convolution, regularization and activation on the input feature map. The C3 module is mainly composed of several Bottleneck structures, and Bottleneck is a A classical residual structure, the input goes through two convolution layers and then performs the Add operation with the original value to complete the residual feature transfer without increasing the output depth. SPP is a spatial pyramid pooling layer. SPP performs three maximum pooling operations of different sizes on the input, and concats the output results to splicing and splicing the output results, so that the output depth of the network is the same as the input depth. The third part is the feature fusion network Neck, which consists of Feature Pyramid Networks (FPN) [24] and Path Aggregation Networks (PAN) [25] structures. The FPN structure combines the category information of high-level large targets To the lower layer, the PAN structure transfers the location information of the low-level large target and the category and location features of the small target upwards. The combination of the two realizes the fusion and complementation of the high-level features and the low-level features, so that the network can obtain more original image information, thereby improving. The detection accuracy of the model for the target. The fourth part is the detection layer Prediction. The three detection layers correspond to the detection and recognition of large, medium and small scale targets respectively. The YOLOv5 algorithm model has four versions: s, m, l and x according to the number of network layers. The change in the number of structural layers is mainly realized by changing the two parameters of depth multiple and width multiple. Among them, YOLOv5s network layer The number is the least, the structure is simpler, and the speed is the fastest.

2.2 Improve YOLOv5s camouflage target detection algorithm

Table 1. Network Structure of CM-YOLOv5s

Num	From	Parameters	Module	Num	From	Parameters	Module
0	−1	3520	Focus	14	−1	131584	Conv
1	−1	18560	Conv	15	−1	0	Upsample
2	−1	18816	C3	16	[-1, 9]	0	Concat
3	−1	1688	CA	17	−1	361984	C3
4	−1	73984	Conv	18	−1	33024	Conv
5	−1	156928	C3	19	−1	0	Upsample
6	−1	3352	CA	20	[-1, 6]	0	Concat
7	−1	295424	Conv	21	−1	90880	C3
8	−1	625152	C3	22	−1	147712	Conv
9	−1	6680	CA	23	[-1, 18]	0	Concat
10	−1	1180672	Conv	24	−1	296448	C3
11	−1	656896	SPP	25	−1	590336	Conv
12	−1	1182720	C3	26	[-1, 14]	0	Concat
13	−1	25648	CA	27	−1	1182720	C3

Although the YOLOv5 detection algorithm is one of the typical and excellent single-stage target detection algorithms, its detection effect on conventional targets is generally better, and it achieves high detection accuracy. However, in the task of camouflage target detection under complex background, because the complex environment background in the picture usually occupies more, so in the process of extracting feature information, it is easy to generate redundant environment background information, which makes it difficult for the network to effectively extract the target feature information. , and with the deepening of the network structure, the target feature information is seriously lost, resulting in poor target detection results. Therefore, this paper proposes a target detection algorithm CM-YOLOv5s for the camouflage characteristics of camouflage targets, such as high concealment, low discrimination and overlapping occlusion, which effectively improves the shortcomings of the original YOLOv5 algorithm and realizes the Accurate identification and rapid positioning of camouflaged targets in complex environments. The network structure of the improved algorithm is shown in Table 1. "From" means the input layer, that is, the output of the previous layer, and "Parameters" indicates the amount of parameters used to describe the complexity of the network.

2.2.1 Coordinate attention mechanism

Because the camouflage target is highly integrated with the surrounding environment background, and the background information usually occupies more in the picture, it is easy to generate a large amount of redundant background information in the process of feature extraction, which leads to the problem of poor detection effect. Therefore, this paper adds the CA attention module to the YOLOv5s backbone feature extraction network Backbone, so that the model can further extract feature information such as orientation perception, so as to more accurately identify and locate camouflaged targets. The CA attention structure is shown in Figure 2.

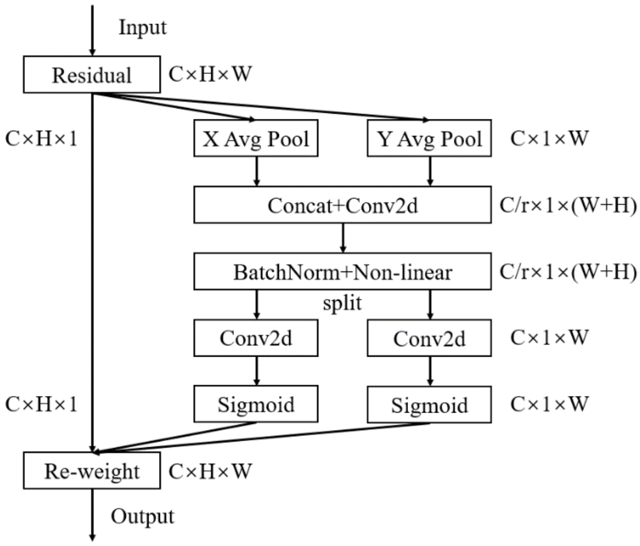

Figure 2. Attention Structure of CA

Assuming X ∈ $R^{C\times H\times W}$ is the input feature map, and then perform the average pooling operations of (H, 1) and (1, W) along the horizontal and vertical coordinates respectively, and the calculation formula is shown in (1) (2).

$$z_c^h(h) = \frac{1}{w}\sum_{0\le i\le w} x_c(h,i) \tag{1}$$

$$z_c^w(w) = \frac{1}{H}\sum_{0\le j\le H} x_c(j,w) \tag{2}$$

The above formula aggregates features along the horizontal and vertical directions to obtain a set of direction-aware feature maps, which is beneficial to the accurate positioning of the network's target of interest. Then use the 1×1 convolution function F_1 to transform, the specific formula is shown in (3).

$$\mathrm{f} = \delta\left(F_1\left([z^h, z^w]\right)\right) \tag{3}$$

In the formula, $f \in R^{C/r\times H\times W}$ represents the intermediate feature map formed by splicing in the horizontal direction and the vertical direction, δ is the splicing function, and r represents the compression ratio. Then divide the feature map f into $f^{h} \in R^{C/r\times H}$ and $f^{w} \in R^{C/r\times W}$, and then use the 1×1 convolution of F_{h} and F_w to convert the feature map f_{h} and f_w into the number of channels of the initial input. The calculation formula is shown in (4) (5).

$$g^{h} = \sigma\left(F_{h}\left(f^{h}\right)\right) \tag{4}$$

$$g^{w} = \sigma\left(F_w(f^{w})\right) \tag{5}$$

In the formula, σ represents the activation function, so the value of g^{h} and g^{w} is between 0 and 1, indicating the importance of direction perception. Finally, the output of the CA module is shown in Equation (6).

$$y_c(i,j) = x_c(i,j) \times g_c^h(i) \times g_c^w(j) \tag{6}$$

With the deepening of the number of neural network layers, the feature information of the target to be detected is gradually lost. Therefore, this paper embeds the CA module into the backbone feature extraction network of the YOLOv5 algorithm to strengthen the attention to the target area of interest in complex environments, effectively improving the original YOLOv5 algorithm network without attention preference and leading to the loss of camouflaged target feature information.

2.2.2 Data Augmentation Strategy

Data augmentation has two benefits: on the one hand, it can enrich the training data and improve the generalization ability and robustness of the model, on the other hand, it does not change the model structure. The Mosaic data enhancement method is mainly used in the YOLOv5 algorithm. In addition, the parameters such as brightness, contrast, saturation, and angle of the image are randomly changed to prevent overfitting.

The idea of Mosaic data enhancement is to use 4 images to be randomly cropped and randomly spliced to combine into one image, and then sent to the network for training, reducing the amount of calculation and speeding up the training speed. The advantages are as follows: First, the local features of the target in the dataset are enriched by random clipping, which can improve the learning ability of the model. The second is to use random stitching to retain all the target features of the image, without discarding the cropped features, and use the stitching method to fully utilize all the features of the image.

In order to improve the network's ability to distinguish occluded and overlapping targets, this paper uses the Mixup method to simulate occlusion and overlapping scenes in camouflaged images in complex environments on the basis of Mosaic data enhancement. The value and label value are weighted respectively to obtain the new image and label value, and then sent to the network for training. During the training process, the labels in the two images are respectively calculated and weighted and summed, and then backpropagated to update the network parameters.

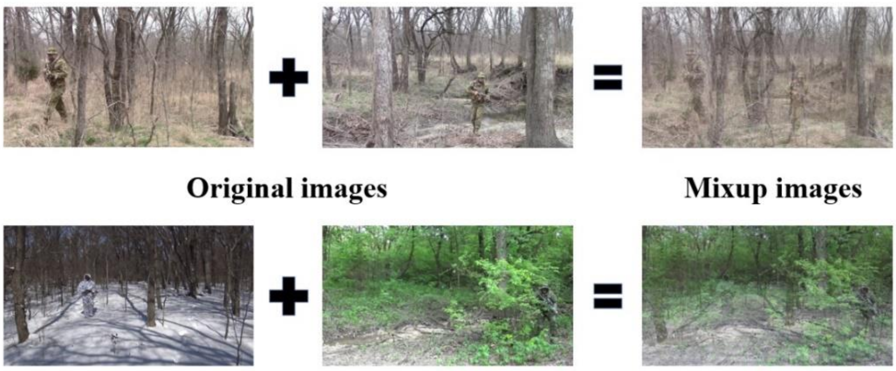

Figure 3. Mixup data augmentation example

The Mixup method can inhibit the overfitting of the neural network, increase the anti-background interference ability of the algorithm, and further improve the generalization ability of the model. The example of Mixup data enhancement in this paper is shown in Figure 3, where "+" represents the weighted summation of the pixel values and label values of the two images, and "=" represents the generation of new images and label values through the Mixup method.

3. Results & Discussion

3.1 Dataset introduction

This paper uses the military camouflage target dataset MCTD for experiments. The data set mainly uses more than 60 military camouflage videos on the Internet in complex field environments as the original material, and initially captures 10,000 high-definition images of camouflage human targets (size of 1280×720) through video frame capture. Camouflage patterns in many countries. The target detection technology based on deep learning mainly relies on three important factors: data, algorithm and computing power, among which high-quality data is the premise for the algorithm to accurately identify the target. Due to the randomness in the video clipping process, it has a great impact on the overall quality of the images in the dataset. Therefore, in order to avoid the influence of the quality of the dataset on the algorithm itself, this paper considers many factors such as multi-directional, multi-angle and target size, attitude, and clarity to screen the dataset images layer by layer, and finally retains relatively high-quality military camouflage target image.

After strict screening, the military camouflage target dataset MCTD contains a total of 1000 high-definition images, each of which contains 1 to 3 camouflaged human targets, and then use the image labeling tool LabelImg to calibrate the human targets in each image one by one. There are two types of human target and environmental background. The following introduces several characteristics of the military camouflage target dataset MCTD constructed in this paper: (1) The target has high concealment and low discrimination; (2) The target scales are different, including large, medium, and small human bodies. Targets; (3) Various target postures, including upright, half-squat, lying down, frontal, back and sideways; (4) Complex environmental backgrounds, including jungles, rainforests, mountains, deserts, snow, and urban ruins 6 kinds of wild environments, and also consider many factors such as season, weather, light and occlusion. Some image samples in the MCTD dataset are shown in Figure 4.

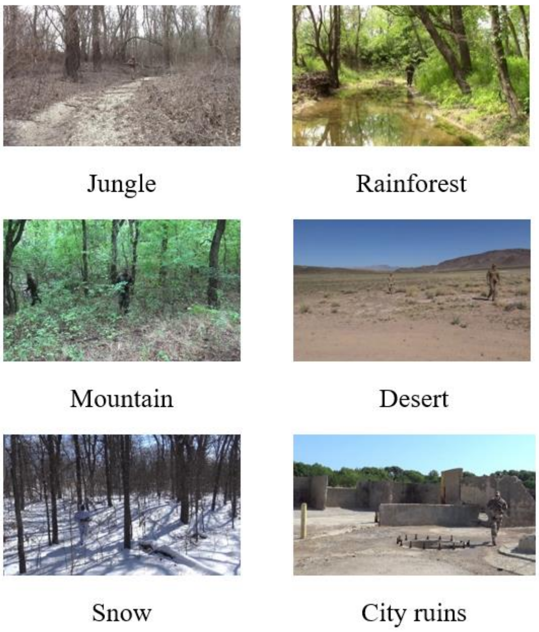

Figure 4. Military Camouflage Target Sample

3.2 Evaluation indicators

In this paper, we choose four items: Precision (P), Recall (R), Mean Average Precision (mAP), Model Size (MB), and FPS (Frames Per Second) The index is used as the evaluation standard of the model. Model inference speed FPS (frame/s) is obtained by averaging the detection time of 200 images in the test set under the server Tesla P100 graphics card environment. The model prediction results include 4 cases: True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN). Among them, the calculation formulas of P, R and mAP are shown in (7)~(10).

$$\mathrm{P} = \frac{TP}{TP + FP} \tag{7}$$

$$\mathrm{R} = \frac{TP}{TP + FN} \tag{8}$$

$$\mathrm{AP} = \frac{1}{n}\sum_{i=1}^{n} P_i = \frac{1}{n}P_1 + \frac{1}{n}P_2 + \cdots + \frac{1}{n}P_n \tag{9}$$

$$mAP = \frac{1}{C}\sum_{k=1}^{C} AP_k \tag{10}$$

In the above formula, the average precision rate (Average precision, AP) refers to the average value of all the accuracies (that is, the area under the PR curve) obtained under all possible values of the recall rate, which is specifically understood as: for a certain type of For samples with n positive examples, all detection results are sorted in descending order of confidence, each additional positive example corresponds to a Precision value (Pi), and the average value of n Pi is obtained to obtain the AP of this class. Mean Precision mAP refers to the mean of AP for each category. In formula (10), C represents the number of classes, and k represents one class.

3.3 Lab environment

The experimental environment of this article is based on the Ubuntu18.04 operating system, the Python version is 3.8, and the deep learning framework is Pytorch1.8.0. The CPU processor is Intel(R) Xeon(R) Gold 6130, the GPU graphics card model is Tesla P100, the video memory size is 15GB, and the memory size is 31GB. CUDA (Compute unified device architecture) and Cudnn (Cuda deep neural network library)) is used for network training to improve the computing power of the computer.

3.4 Network training

Before training, the military camouflage target data set MCTD is randomly divided into training set, validation set and test set with a ratio of 6:2:2. In order to fully train the improved network model and prevent the neural network from overfitting, a variety of data enhancement methods are used to expand the data set during training, including color transformation, left-right flip, translation zoom, Mosaic, and Mixup.

The number of training is set to 100, 32 military camouflage target images are processed in each batch, the size of the input image is 640×640, the initial learning rate is 0.0001, the learning rate momentum and weight decay coefficient are set to 0.937, 0.0005, respectively, and the Mixup parameter is set to 0.1, the optimization strategy selects the Adam [26] (Adaptive Moment Estimation) optimizer.

3.5 Experiment and result analysis

In this paper, two groups of experiments are designed for verification, which are different improved partial ablation experiments and different algorithm model comparison experiments. The specific content and results of this experiment will be introduced in detail below.

3.5.1 Ablation experiment

In order to test the effectiveness of the improved algorithm, ablation comparison experiments were carried out, and three groups of network models with different improved parts were verified. All experiments were carried out on the military camouflage target dataset MCTD. The results of ablation experiments are shown in Table 2, where "√" indicates that the corresponding improved method is used in the network model, and "×" indicates that the corresponding improved method is not used in the network model.

Table 2. Comparison of Ablation Experiment Results

Model	CA	Mixup	P/%	R/%	$mAP_{0.5}$/%	Size/MB
YOLOv5s	×	×	94.1	83.3	90.3	14.4
YOLOv5s	√	×	95.5	88.9	92.8	14.5
YOLOv5s	×	√	94.9	85.7	93.4	14.4
YOLOv5s	√	√	95.9	87.1	94.1	14.5

It can be seen from the data in Table 3 that the CA attention module is introduced into the backbone feature extraction network Backbone, which improves the network's ability to capture feature information such as target position and direction, and improves the algorithm's accuracy P, recall rate R and The average precision and mean mAP are increased by 1.4%, 5.6%, and 2.5% respectively, which effectively solves the problem of loss of camouflage target feature information due to the lack of attention preference in the network, and improves the detection effect of camouflage targets. The Mixup data enhancement strategy is used to simulate Training in overlapping occlusion scenes improves the network's learning ability for complex samples, and also makes the network model's generalization ability stronger. As can be seen from the table, this method makes the algorithm's precision P, recall rate R and average precision. The mean mAP was also improved by 0.8%, 2.4%, and 3.1%, respectively. Compared with the original algorithm, the combination of the two has greatly improved various evaluation indicators, which not only improves the problem that the original network is difficult to fully extract and integrate multi-layer features, but also effectively alleviates the problem of overlapping occlusion in complex environments. This paper improves the effectiveness of the algorithm. In summary, the improved algorithm achieves accurate detection and recognition of camouflaged targets in complex environments.

3.5.2 Comparison of different algorithm models

In order to further verify the detection effect of the improved method in this paper, the improved algorithm is compared with several mainstream object detection models: RetinaNet, YOLOX [27], YOLOv5s, YOLOv5m and YOLOv5l. The data set and platform configuration conditions used in the experiment are the same, and the performance comparison results are shown in Table 3. The bold font is the optimal value of the project.

Table 3. Detection performance of different models

Model	$mAP_{0.5}$/%	FPS	Size/MB
Retinanet	81.4	6.4	138
YOLOX	88.7	13.4	34.3
YOLOv5s	90.3	**125**	**14.4**
YOLOv5m	92.6	45.5	40.4
YOLOv5l	93.4	43.5	93.7
CM-YOLOv5s	**94.1**	81.1	14.5

From the data in Table 4, it can be seen that the model size of the improved algorithm is only 14.5MB, which is only 0.1MB higher than the original YOLOv5s algorithm, and the detection speed FPS is reduced by 43.9. However, the detection accuracy of the improved model for military camouflage targets is improved by 3.8 percent. Compared with the four excellent algorithms such as RetinaNet, YOLOX, YOLOv5m and YOLOv5l, the CM-YOLOv5s algorithm model is more lightweight, and is superior to the appeal algorithm in terms of detection accuracy and inference speed. The detection performance is the best.

3.6 Test results

Figure 5 shows the detection results of five algorithms in this paper, CM-YOLOv5s, YOLOv5s, YOLOX, YOLOv5m, and YOLOv5l, on camouflaged personnel under typical complex backgrounds. From the detection results in the figure, compared with the original model YOLOv5s and the current advanced Anchor free-based high-performance detector YOLOX, it is obvious that the improved algorithm has higher detection accuracy and can effectively capture the characteristic information of camouflage targets, with better robustness. Compared with the YOLOv5m and YOLOv5l models with deeper network structure, the improved algorithm can still show relatively better detection performance. The detection results show that the algorithm in this paper has strong generalization ability for the detection task of camouflage targets in complex environments, and can effectively identify and accurately locate camouflage targets.

Figure 5. Comparison of detection results of different models

4. Conclusion

This paper proposes a target detection algorithm CM-YOLOv5s for camouflage characteristics, aiming at the problem that the target and the complex environment background are highly integrated and the discrimination is low in the camouflage target detection task, resulting in poor detection effect. The algorithm adds the CA attention module to the YOLOv5s backbone feature extraction network, which weakens the complex background information in the picture, enhances the network's feature extraction of the camouflaged target area, and effectively improves the network's anti-background interference ability. Secondly, the Mixup data enhancement strategy is adopted. Simulating overlapping occlusion scenarios for training improves the network's learning ability for complex samples, and also makes the network model's generalization ability stronger. The experimental results show that the detection effect of the CM-YOLOv5s network model is better. Compared with the original YOLOv5s algorithm model, the precision rate, recall rate and average precision mAP are increased by 1.8%, 3.8%, and 3.8%

respectively. Accurate detection, identification and positioning of camouflaged targets in complex environments are achieved.

References

[1] Wang Yang, Cao Tieyong, Yang Jibin, et al. Research on camouflage target detection technology based on YOLO v5 algorithm [J]. Computer Science, 2021,48(10): 226-232.
[2] Liang Xinyu, Lin Haokun, Yang Hui, et al. Construction of a dataset for semantic segmentation of camouflage target images [J]. Advances in Lasers and Optoelectronics, 2021,58(04): 214-220.
[3] HEARST M A, DUMAIS S T, OSMAN E, et al. Support vector machines[J]. ieee intelligent systems & their applications, 1998,13(4): 18-28.
[4] SCHWENK H, BENGIO Y. AdaBoosting Neural Networks: Application to on-line Character Recognition: International Conference on Artificial Neural Networks[Z]. 1997967-972.
[5] FELZENSZWALB P F, GIRSHICK R B, MCALLESTER D, et al. Object Detection with Discriminatively Trained Part-Based Models[J]. ieee transactions on pattern analysis and machine intelligence, 2010,32(9): 1627-1645.
[6] BREIMAN L. Random Forests[Z]. 2001: 45, 5-32.
[7] GIRSHICK R, DONAHUE J, DARRELL T, et al. Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation: Computer Vision and Pattern Recognition[Z]. 2014580-587.
[8] GIRSHICK R. Fast R-CNN: International Conference on Computer Vision[Z]. 20151440-1448.
[9] Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks[J]. 2017.
[10] LIU W, ANGUELOV D, ERHAN D, et al. SSD: Single Shot MultiBox Detector: European Conference on Computer Vision[Z]. 201621-37.
[11] TAN M, LE Q V. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks[J]. 2019.
[12] LIN T, GOYAL P, GIRSHICK R, et al. Focal Loss for Dense Object Detection: International Conference on Computer Vision[Z]. 20172999-3007.
[13] REDMON J, DIVVALA S, GIRSHICK R, et al. You Only Look Once: Unified, Real-Time Object Detection: Computer Vision and Pattern Recognition[Z]. 2016779-788
[14] REDMON J, FARHADI A. YOLO9000: Better, Faster, Stronger: Computer Vision and Pattern Recognition[Z]. 20176517-6525.
[15] REDMON J, FARHADI A. YOLOv3: An Incremental Improvement.[Z]. 2018.
[16] BOCHKOVSKIY A, WANG C, LIAO H M. YOLOv4: Optimal Speed and Accuracy of Object Detection[Z]. 2020.
[17] Wang Zhi, Chen Ping, Pan Jinxiao. Object detection in complex background based on deep learning [J]. Journal of Chongqing University of Technology (Natural Science), 2018,32(04): 171-176.
[18] Yu Bowen, Lv Ming. Improved YOLOv3 algorithm and its application in military target detection [J]. Journal of Ordnance Engineering: 1-10.
[19] Deng Xiaotong, Cao Tieyong, Fang Zheng, et al. Research on improved RetinaNet camouflaged person detection method [J]. Computer Engineering and Applications, 2021,57(05): 190-196.
[20] HOU Q, ZHOU D, FENG J. Coordinate Attention for Efficient Mobile Network Design[J]. 2021.
[21] ZHANG H, CISSE M, DAUPHIN Y N, et al. mixup: Beyond Empirical Risk Minimization[J]. 2017.
[22] ZIHAO L Y L B. Research on the Use of YOLOv5 Object Detection Algorithm in Mask Wearing Recognition[Z]. 2020: 6, 276-284.
[23] HE K, ZHANG X, REN S, et al. Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition[J]. ieee transactions on pattern analysis and machine intelligence, 2015,37(9): 1904-1916.
[24] LIN T, DOLLAR P, GIRSHICK R, et al. Feature Pyramid Networks for Object Detection: Computer Vision and Pattern Recognition[Z]. 2017936-944.

[25] CHEN Y, WANG Y, ZHANG Y, et al. PANet: A Context Based Predicate Association Network for Scene Graph Generation: International Conference on Multimedia and Expo[Z]. 2019508-513.
[26] KINGMA D, BA J. Adam: A Method for Stochastic Optimization[J]. Computer Science, 2014.
[27] GE Z, LIU S, WANG F, et al. YOLOX: Exceeding YOLO Series in 2021[J]. 2021.